

MTL411: Functional Analysis

Lecture 17: Dual Spaces and ℓ^p Spaces

In-Class Notes

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1 Preliminaries

1.1 Dual Spaces

Let X be a normed linear space over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$.

- The **continuous dual** of X is

$$X' = B(X, \mathbb{K}) = \{f : X \rightarrow \mathbb{K} \mid f \text{ is linear and bounded}\}.$$

- The **algebraic dual** is

$$X^* = L(X, \mathbb{K}) = \{f : X \rightarrow \mathbb{K} \mid f \text{ is linear}\}.$$

- Every bounded linear functional is continuous, so

$$X' \subset X^*.$$

1.2 Finite Dimensional Case

Let $X = \mathbb{R}^n$ with standard basis $\{e_1, \dots, e_n\}$.

Any $x \in X$ can be written as

$$x = \sum_{j=1}^n x(j)e_j.$$

Let $f \in X'$. Then by linearity,

$$f(x) = \sum_{j=1}^n x(j)f(e_j).$$

Define $y = (y(1), \dots, y(n)) \in \mathbb{R}^n$ by

$$y(j) = f(e_j).$$

Thus,

$$f(x) = \sum_{j=1}^n x(j)y(j).$$

1.3 Isomorphism with $\mathbb{R}^n$

Define a map

$$T : (\mathbb{R}^n)' \rightarrow \mathbb{R}^n, \quad T(f) = (f(e_1), \dots, f(e_n)).$$

Then:

- T is linear,
- T is bijective,

hence $(\mathbb{R}^n)' \cong \mathbb{R}^n$.

2 ℓ^p Spaces and Duality

2.1 Basic Setup

Let $1 \leq p < \infty$ and let q be such that

$$\frac{1}{p} + \frac{1}{q} = 1.$$

Let $x \in \ell^p$ and $y \in \ell^q$.

By Hölder's inequality,

$$\sum_{j=1}^{\infty} |x(j)y(j)| \leq \|x\|_p \|y\|_q,$$

so the series converges.

2.2 Canonical Functional

For each $y \in \ell^q$, define

$$f_y : \ell^p \rightarrow \mathbb{K}, \quad f_y(x) = \sum_{j=1}^{\infty} x(j)y(j).$$

Then:

- f_y is linear,
- by Hölder inequality,

$$|f_y(x)| \leq \|x\|_p \|y\|_q,$$

so $f_y \in (\ell^p)'$.

2.3 Natural Map

Define

$$T : \ell^q \rightarrow (\ell^p)', \quad T(y) = f_y.$$

Then T is linear.

3 Theorem: Surjective Linear Isometry

Theorem 3.1. Let $1 \leq p < \infty$ and let q be such that $\frac{1}{p} + \frac{1}{q} = 1$. Define

$$T : \ell^q \rightarrow (\ell^p)', \quad T(y) = f_y, \quad f_y(x) = \sum_{j=1}^{\infty} x(j)y(j).$$

Then T is a surjective linear isometry. That is:

1. T is linear,
2. $\|T(y)\| = \|y\|_q$ for all $y \in \ell^q$,
3. T is onto.

Proof. Step 1: Linearity.

For $y_1, y_2 \in \ell^q$ and $\alpha \in \mathbb{K}$,

$$T(y_1 + \alpha y_2)(x) = \sum x(j)(y_1(j) + \alpha y_2(j)) = f_{y_1}(x) + \alpha f_{y_2}(x).$$

Thus T is linear.

Step 2: $\|T(y)\| \leq \|y\|_q$.

By Hölder's inequality,

$$|f_y(x)| \leq \|x\|_p \|y\|_q.$$

Taking supremum over $\|x\|_p \leq 1$,

$$\|T(y)\| = \|f_y\| \leq \|y\|_q.$$

Step 3: $\|T(y)\| \geq \|y\|_q$.

This is the key step.

Define, for $y \in \ell^q$,

$$x(j) = \begin{cases} \frac{|y(j)|^{q-1} \operatorname{sgn}(y(j))}{\|y\|_q^{q-1}}, & y(j) \neq 0, \\ 0, & y(j) = 0. \end{cases}$$

Then one can verify that $x \in \ell^p$ and $\|x\|_p = 1$.

Now compute:

$$f_y(x) = \sum x(j)y(j) = \frac{1}{\|y\|_q^{q-1}} \sum |y(j)|^q = \frac{\|y\|_q^q}{\|y\|_q^{q-1}} = \|y\|_q.$$

Hence,

$$\|T(y)\| \geq |f_y(x)| = \|y\|_q.$$

Combining with Step 2,

$$\|T(y)\| = \|y\|_q.$$

Step 4: Surjectivity.

Let $f \in (\ell^p)'$. Define

$$y(j) := f(e_j),$$

where $\{e_j\}$ is the standard basis.

We claim that $y \in \ell^q$.

For each n , define

$$x_n(j) = \begin{cases} \frac{|y(j)|^{q-1} \operatorname{sgn}(y(j))}{(\sum_{k=1}^n |y(k)|^q)^{\frac{q-1}{q}}}, & j \leq n, \\ 0, & j > n. \end{cases}$$

Then $\|x_n\|_p = 1$, so

$$|f(x_n)| \leq \|f\|.$$

But,

$$f(x_n) = \sum_{j=1}^n x_n(j)y(j) = \left(\sum_{j=1}^n |y(j)|^q \right)^{\frac{1}{q}}.$$

Thus,

$$\left(\sum_{j=1}^n |y(j)|^q \right)^{\frac{1}{q}} \leq \|f\|.$$

Taking $n \rightarrow \infty$, we get $y \in \ell^q$ and

$$\|y\|_q \leq \|f\|.$$

Now for any $x \in \ell^p$,

$$f(x) = \sum x(j)f(e_j) = \sum x(j)y(j) = f_y(x).$$

Thus $f = T(y)$, so T is surjective. □

4 Boundedness of Coordinate Functionals

Lemma 4.1. *Let $1 \leq p < \infty$ and let $f \in (\ell^p)'$. Define*

$$y(j) := f(e_j), \quad j \in \mathbb{N}.$$

Then $y \in \ell^\infty$ and

$$\|y\|_\infty \leq \|f\|.$$

Proof. For each j , consider the standard basis vector $e_j \in \ell^p$. Since $\|e_j\|_p = 1$, we have

$$|y(j)| = |f(e_j)| \leq \|f\| \|e_j\|_p = \|f\|.$$

Thus,

$$\sup_j |y(j)| \leq \|f\|,$$

so $y \in \ell^\infty$ and $\|y\|_\infty \leq \|f\|$. □

5 Surjectivity of the Map T

Theorem 5.1. *Let $1 \leq p < \infty$ and let q be such that $\frac{1}{p} + \frac{1}{q} = 1$. Let*

$$T : \ell^q \rightarrow (\ell^p)', \quad T(y) = f_y.$$

Then T is surjective.

Proof. Let $f \in (\ell^p)'$.

Define a sequence $y = (y(j))$ by

$$y(j) := f(e_j), \quad j \in \mathbb{N}.$$

Step 1: Show $y \in \ell^q$.

For each $n \in \mathbb{N}$, define

$$x_n(j) = \begin{cases} \frac{|y(j)|^{q-1} \operatorname{sgn}(y(j))}{y(j)} & \text{if } j \leq n, y(j) \neq 0, \\ 0 & \text{otherwise.} \end{cases}$$

(Equivalently, your notes define x_n so that the sum below works — this is the same idea.)
Then,

$$\sum_{j=1}^n |y(j)|^q = \sum_{j=1}^n x_n(j) y(j) = f(x_n).$$

Using boundedness of f ,

$$|f(x_n)| \leq \|f\| \|x_n\|_p.$$

Hence,

$$\sum_{j=1}^n |y(j)|^q \leq \|f\| \|x_n\|_p.$$

Now compute $\|x_n\|_p$:

$$\|x_n\|_p^p = \sum_{j=1}^n |x_n(j)|^p = \sum_{j=1}^n |y(j)|^q.$$

Thus,

$$\|x_n\|_p = \left(\sum_{j=1}^n |y(j)|^q \right)^{1/p}.$$

Substituting back,

$$\sum_{j=1}^n |y(j)|^q \leq \|f\| \left(\sum_{j=1}^n |y(j)|^q \right)^{1/p}.$$

Divide both sides:

$$\left(\sum_{j=1}^n |y(j)|^q \right)^{1 - \frac{1}{p}} \leq \|f\|.$$

Since $1 - \frac{1}{p} = \frac{1}{q}$,

$$\left(\sum_{j=1}^n |y(j)|^q \right)^{1/q} \leq \|f\|.$$

Taking $n \rightarrow \infty$, we get

$$y \in \ell^q \quad \text{and} \quad \|y\|_q \leq \|f\|.$$

Step 2: Show $f = f_y$.

Let $x \in \ell^p$. Then

$$x = \sum_{j=1}^{\infty} x(j)e_j.$$

Using linearity and continuity,

$$f(x) = \sum_{j=1}^{\infty} x(j)f(e_j) = \sum_{j=1}^{\infty} x(j)y(j).$$

Thus,

$$f = f_y = T(y).$$

Hence T is surjective. □

6 Dense Subspaces and Duals

6.1 Preliminary Observations

Let $X_0 \subset X$ be a dense subspace of a normed linear space X .

Then, intuitively,

$$X' \cong X'_0,$$

i.e., every continuous linear functional on X is uniquely determined by its restriction to X_0 .

Examples:

- c_{00} (finite support sequences) is dense in ℓ^p for $1 \leq p < \infty$.
- Hence, information about $(\ell^p)'$ can be recovered from functionals defined on c_{00} .
- In particular,

$$(\ell^1)' \cong \ell^\infty.$$

6.2 Dense Subspace Lemma

Lemma 6.1. *Let X_0 be a dense subspace of a normed linear space X . Define*

$$F : X' \rightarrow X'_0, \quad F(f) = f|_{X_0}.$$

Then F is a linear isometry. In particular,

$$\|F(f)\| = \|f\|.$$

Proof. **Step 1: Linearity.**

For $f, g \in X'$ and $\alpha \in \mathbb{K}$,

$$F(f + \alpha g) = (f + \alpha g)|_{X_0} = f|_{X_0} + \alpha g|_{X_0},$$

so F is linear.

Step 2: $\|F(f)\| \leq \|f\|$.

For any $x \in X_0$,

$$|F(f)(x)| = |f(x)| \leq \|f\| \|x\|.$$

Taking supremum over $\|x\| \leq 1$ in X_0 ,

$$\|F(f)\| \leq \|f\|.$$

Step 3: $\|F(f)\| \geq \|f\|$.

Let $x \in X$ with $\|x\| = 1$. Since X_0 is dense in X , there exists a sequence $(x_n) \subset X_0$ such that

$$x_n \rightarrow x.$$

By continuity of f ,

$$f(x_n) \rightarrow f(x).$$

Thus,

$$|f(x)| = \lim_{n \rightarrow \infty} |f(x_n)| \leq \|F(f)\|.$$

Taking supremum over all $\|x\| \leq 1$,

$$\|f\| \leq \|F(f)\|.$$

Combining both inequalities,

$$\|F(f)\| = \|f\|.$$

□

7 Applications of Dense Subspace Lemma

7.1 Density of c_{00} in ℓ^p

Lemma 7.1. For $1 \leq p < \infty$, the space c_{00} (sequences with finite support) is dense in ℓ^p .

Proof. Let $x = (x(j)) \in \ell^p$. Define

$$x_n = (x(1), x(2), \dots, x(n), 0, 0, \dots).$$

Then $x_n \in c_{00}$ and

$$\|x - x_n\|_p^p = \sum_{j=n+1}^{\infty} |x(j)|^p \rightarrow 0.$$

Thus $x_n \rightarrow x$ in ℓ^p , so c_{00} is dense in ℓ^p .

□

7.2 Dual of ℓ^p

Using the density of c_{00} in ℓ^p and the previous theorem,

$$(\ell^p)' \cong \ell^q, \quad 1 \leq p < \infty.$$

7.3 Dual of c_0

Theorem 7.2. *The dual of c_0 is ℓ^1 , i.e.,*

$$(c_0)' \cong \ell^1.$$

Proof. Let $y \in \ell^1$. Define

$$f_y(x) = \sum_{j=1}^{\infty} x(j)y(j), \quad x \in c_0.$$

Then f_y is linear and

$$|f_y(x)| \leq \|x\|_{\infty} \|y\|_1,$$

so $f_y \in (c_0)'$.

Thus we get a map

$$T : \ell^1 \rightarrow (c_0)', \quad T(y) = f_y.$$

One can show:

- T is linear,
- $\|T(y)\| = \|y\|_1$,
- T is surjective (using density of c_{00} in c_0).

Hence $(c_0)' \cong \ell^1$. □

7.4 Dual of c

Theorem 7.3. *Let c be the space of convergent sequences with the sup norm. Then*

$$c' \cong \ell^1 \oplus \mathbb{K}.$$

Proof. Let $x \in c$. Then $x(n) \rightarrow \alpha$ as $n \rightarrow \infty$.

Define

$$e_0 = (1, 1, 1, \dots).$$

Then we can write

$$x = u + \alpha e_0,$$

where

$$u = x - \alpha e_0 = (x(1) - \alpha, x(2) - \alpha, \dots).$$

Since $x(n) \rightarrow \alpha$, we have $u \in c_0$.

Thus,

$$c = c_0 \oplus \text{span}\{e_0\}.$$

Now let $f \in c'$. Then for $x = u + \alpha e_0$,

$$f(x) = f(u + \alpha e_0) = f(u) + \alpha f(e_0).$$

Since $u \in c_0$ and $(c_0)' \cong \ell^1$, there exists $y \in \ell^1$ such that

$$f(u) = \sum_{j=1}^{\infty} u(j)y(j).$$

Thus,

$$f(x) = \alpha f(e_0) + \sum_{j=1}^{\infty} u(j)y(j).$$

Substitute $u(j) = x(j) - \alpha$:

$$f(x) = \alpha f(e_0) + \sum_{j=1}^{\infty} (x(j) - \alpha)y(j).$$

Expanding,

$$f(x) = \alpha f(e_0) + \sum_{j=1}^{\infty} x(j)y(j) - \alpha \sum_{j=1}^{\infty} y(j).$$

Rewriting,

$$f(x) = \sum_{j=1}^{\infty} x(j)y(j) + \alpha \left(f(e_0) - \sum_{j=1}^{\infty} y(j) \right).$$

Let

$$\beta = f(e_0) - \sum_{j=1}^{\infty} y(j).$$

Thus,

$$f(x) = \sum_{j=1}^{\infty} x(j)y(j) + \alpha\beta.$$

Hence every $f \in c'$ corresponds to a pair $(y, \beta) \in \ell^1 \oplus \mathbb{K}$.

Conversely, any such pair defines a bounded linear functional.

Therefore,

$$c' \cong \ell^1 \oplus \mathbb{K}.$$

□

MTL411: Functional Analysis

Lecture 18: Hahn–Banach Theorem

In-Class Notes

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1 Hahn–Banach Theorem (Real Case)

Theorem 1.1 (Hahn–Banach Theorem, Real Case). *Let X be a real normed linear space, and let $M \subset X$ be a subspace. Suppose $f : M \rightarrow \mathbb{R}$ is a continuous linear functional. Then there exists an extension $\tilde{f} : X \rightarrow \mathbb{R}$ such that:*

1. $\tilde{f}|_M = f$,
2. $\|\tilde{f}\| = \|f\|$.

Proof. Without loss of generality, assume $\|f\| = 1$ (otherwise consider $f/\|f\|$).

Let $z \in X \setminus M$. We aim to extend f to the larger subspace

$$M_1 := \{x + \alpha z : x \in M, \alpha \in \mathbb{R}\}.$$

Define a function $h : M_1 \rightarrow \mathbb{R}$ by

$$h(x + \alpha z) = f(x) + \alpha c,$$

where $c \in \mathbb{R}$ is to be determined.

We want h to satisfy $\|h\| \leq 1$, i.e.,

$$|h(x + \alpha z)| \leq \|x + \alpha z\| \quad \text{for all } x \in M, \alpha \in \mathbb{R}.$$

This is equivalent to requiring:

$$|f(x) + \alpha c| \leq \|x + \alpha z\|.$$

For $\alpha > 0$, divide by α :

$$\left| \frac{f(x)}{\alpha} + c \right| \leq \left\| \frac{x}{\alpha} + z \right\|.$$

Let $y = \frac{x}{\alpha} \in M$, then:

$$|f(y) + c| \leq \|y + z\|.$$

This gives:

$$-f(y) - \|y + z\| \leq c \leq \|y + z\| - f(y).$$

Thus, c must satisfy:

$$\sup_{y \in M} (-\|y + z\| - f(y)) \leq c \leq \inf_{y \in M} (\|y + z\| - f(y)).$$

We now show that the interval is nonempty. For any $y_1, y_2 \in M$,

$$f(y_1 - y_2) \leq \|y_1 - y_2\| \leq \|y_1 + z\| + \|y_2 + z\|.$$

Rearranging,

$$-f(y_2) - \|y_2 + z\| \leq \|y_1 + z\| - f(y_1).$$

Taking supremum over y_2 and infimum over y_1 , we get:

$$\sup_{y \in M} (-\|y + z\| - f(y)) \leq \inf_{y \in M} (\|y + z\| - f(y)).$$

Hence, such a c exists, and h is a norm-preserving extension of f to M_1 .

Now consider the collection $\mathcal{P}$ of all pairs (N, g) where:

- $M \subset N \subset X$ is a subspace,
- $g : N \rightarrow \mathbb{R}$ is linear,
- $g|_M = f$ and $\|g\| = \|f\|$.

Partially order $\mathcal{P}$ by extension: $(N_1, g_1) \leq (N_2, g_2)$ if $N_1 \subset N_2$ and $g_2|_{N_1} = g_1$.

By Zorn's Lemma, there exists a maximal element $(\tilde{N}, \tilde{f})$.

If $\tilde{N} \neq X$, then we can extend $\tilde{f}$ further by the previous step, contradicting maximality. Hence $\tilde{N} = X$.

Therefore, $\tilde{f}$ is the desired extension. □

1.1 Zorn's Lemma and Order Structures

Definition 1.2 (Partially Ordered Set (Poset)). A set P together with a relation $\leq$ is called a **partially ordered set** if for all $x, y, z \in P$:

1. (Reflexive) $x \leq x$,
2. (Antisymmetric) $x \leq y$ and $y \leq x$ imply $x = y$,
3. (Transitive) $x \leq y$ and $y \leq z$ imply $x \leq z$.

Definition 1.3 (Totally Ordered Set). A partially ordered set $(P, \leq)$ is called **totally ordered** (or a chain) if for every $x, y \in P$, either $x \leq y$ or $y \leq x$.

Lemma 1.4. *The collection*

$$\mathcal{P} = \{(N, g) : M \subset N \subset X, g : N \rightarrow \mathbb{R} \text{ linear}, g|_M = f, \|g\| = \|f\|\}$$

with the relation

$$(N_1, g_1) \leq (N_2, g_2) \iff N_1 \subset N_2 \text{ and } g_2|_{N_1} = g_1$$

is a partially ordered set.

Proof. We verify the three properties:

Reflexivity: For any $(N, g) \in \mathcal{P}$, clearly $N \subset N$ and $g|_N = g$, so $(N, g) \leq (N, g)$.

Antisymmetry: If $(N_1, g_1) \leq (N_2, g_2)$ and $(N_2, g_2) \leq (N_1, g_1)$, then

$$N_1 \subset N_2 \quad \text{and} \quad N_2 \subset N_1 \implies N_1 = N_2.$$

Also,

$$g_2|_{N_1} = g_1 \quad \text{and} \quad g_1|_{N_2} = g_2,$$

so $g_1 = g_2$. Hence $(N_1, g_1) = (N_2, g_2)$.

Transitivity: If $(N_1, g_1) \leq (N_2, g_2)$ and $(N_2, g_2) \leq (N_3, g_3)$, then

$$N_1 \subset N_2 \subset N_3 \implies N_1 \subset N_3,$$

and

$$g_3|_{N_1} = (g_3|_{N_2})|_{N_1} = g_2|_{N_1} = g_1.$$

Thus $(N_1, g_1) \leq (N_3, g_3)$.

Hence $\mathcal{P}$ is a poset. □

Theorem 1.5 (Zorn's Lemma). *Let $(P, \leq)$ be a partially ordered set. If every totally ordered subset (chain) of P has an upper bound in P , then P has a maximal element.*

Remark 1.6. In the proof of the Hahn–Banach theorem, one shows that every chain in $\mathcal{P}$ has an upper bound (constructed by taking unions of domains and consistent extensions). Hence, by Zorn's Lemma, $\mathcal{P}$ admits a maximal element.

2 Hahn–Banach Theorem (Complex Case)

Lemma 2.1. *Let X be a complex normed linear space. A function $f : X \rightarrow \mathbb{C}$ is a continuous linear functional if and only if there exists a unique real-valued continuous linear functional $u : X \rightarrow \mathbb{R}$ such that*

$$f(x) = u(x) - i u(ix) \quad \text{for all } x \in X,$$

and moreover,

$$\|f\| = \|u\|.$$

Proof. (Only if part)

Let $f : X \rightarrow \mathbb{C}$ be a continuous linear functional. Define

$$u(x) := \operatorname{Re} f(x).$$

Then $u : X \rightarrow \mathbb{R}$ is real-linear and continuous, and clearly

$$|u(x)| \leq |f(x)| \leq \|f\| \|x\|,$$

so $\|u\| \leq \|f\|$.

Now observe that for any $x \in X$,

$$f(x) = \operatorname{Re} f(x) + i \operatorname{Im} f(x).$$

Also,

$$\operatorname{Im} f(x) = -\operatorname{Re} f(ix),$$

since

$$f(ix) = if(x) \implies \operatorname{Re} f(ix) = -\operatorname{Im} f(x).$$

Thus,

$$f(x) = u(x) - i u(ix).$$

To prove $\|f\| \leq \|u\|$, fix $x \in X$, $x \neq 0$, and define

$$\alpha := \frac{\overline{f(x)}}{|f(x)|} \quad \text{so that } |\alpha| = 1.$$

Then

$$f(\alpha x) = \alpha f(x) = |f(x)|.$$

Taking real parts,

$$|f(x)| = \operatorname{Re} f(\alpha x) = u(\alpha x).$$

Hence,

$$|f(x)| \leq \|u\| \|\alpha x\| = \|u\| \|x\|.$$

Taking supremum over $\|x\| = 1$, we get $\|f\| \leq \|u\|$.

Thus $\|f\| = \|u\|$.

(If part)

Conversely, let $u : X \rightarrow \mathbb{R}$ be a real continuous linear functional. Define

$$f(x) := u(x) - i u(ix).$$

Then f is complex-linear. Indeed, for $\lambda = a + ib \in \mathbb{C}$,

$$f(\lambda x) = u(ax - b(ix)) - i u(i(ax - b(ix))) = au(x) + bu(ix) - i(au(ix) - bu(x)) = (a + ib)(u(x) - i u(ix)) = \lambda f(x).$$

Also,

$$|f(x)| \leq |u(x)| + |u(ix)| \leq 2\|u\| \|x\|,$$

so f is continuous.

Finally, the representation shows uniqueness of u as $u(x) = \operatorname{Re} f(x)$. □

Theorem 2.2 (Hahn–Banach Theorem, Complex Case). *Let X be a complex normed linear space, $M \subset X$ a subspace, and $f : M \rightarrow \mathbb{C}$ a continuous linear functional. Then there exists an extension $\tilde{f} : X \rightarrow \mathbb{C}$ such that*

1. $\tilde{f}|_M = f$,
2. $\|\tilde{f}\| = \|f\|$.

Proof. Let $u = \operatorname{Re} f$. Then $u : M \rightarrow \mathbb{R}$ is a real continuous linear functional with $\|u\| = \|f\|$.

By the real Hahn–Banach theorem, there exists an extension $\tilde{u} : X \rightarrow \mathbb{R}$ such that

$$\tilde{u}|_M = u, \quad \|\tilde{u}\| = \|u\|.$$

Define $\tilde{f} : X \rightarrow \mathbb{C}$ by

$$\tilde{f}(x) = \tilde{u}(x) - i \tilde{u}(ix).$$

Then $\tilde{f}$ is a complex linear functional extending f . Indeed, for $x \in M$,

$$\tilde{f}(x) = \tilde{u}(x) - i \tilde{u}(ix) = u(x) - i u(ix) = f(x).$$

Moreover, by the lemma,

$$\|\tilde{f}\| = \|\tilde{u}\| = \|u\| = \|f\|.$$

Thus $\tilde{f}$ is the desired extension. □

3 Corollaries

3.1 Corollary 1: Norm Attaining Functional

Corollary 3.1. *Let X be a normed linear space and let $x_0 \in X$, $x_0 \neq 0$. Then there exists a continuous linear functional $f \in X^*$ such that*

$$\|f\| = 1 \quad \text{and} \quad f(x_0) = \|x_0\|.$$

Proof. Let $M = \text{span}\{x_0\} = \{\alpha x_0 : \alpha \in \mathbb{R}\}$.

Define $f_0 : M \rightarrow \mathbb{R}$ by

$$f_0(\alpha x_0) = \alpha \|x_0\|.$$

Step 1: f_0 is linear. For $\alpha, \beta \in \mathbb{R}$,

$$f_0(\alpha x_0 + \beta x_0) = f_0((\alpha + \beta)x_0) = (\alpha + \beta)\|x_0\| = f_0(\alpha x_0) + f_0(\beta x_0).$$

Step 2: Compute the norm of f_0 . For any $\alpha \in \mathbb{R}$,

$$|f_0(\alpha x_0)| = |\alpha| \|x_0\| = \|\alpha x_0\|.$$

Thus,

$$\frac{|f_0(x)|}{\|x\|} = 1 \quad \text{for all } x \in M, x \neq 0,$$

so $\|f_0\| = 1$.

Step 3: Apply Hahn–Banach. By the Hahn–Banach theorem (real case), f_0 extends to a continuous linear functional $f : X \rightarrow \mathbb{R}$ such that

$$f|_M = f_0 \quad \text{and} \quad \|f\| = \|f_0\| = 1.$$

Step 4: Evaluate at x_0 . Since $x_0 \in M$,

$$f(x_0) = f_0(x_0) = \|x_0\|.$$

This completes the proof. □

3.2 Corollary 2: Dual Characterization of Norm

Corollary 3.2. *Let X be a normed linear space. Then for every $x \in X$,*

$$\|x\| = \sup_{\substack{f \in X^* \\ f \neq 0}} \frac{|f(x)|}{\|f\|} = \sup_{\|f\|=1} |f(x)|.$$

Proof. **Step 1: Show** $\sup_{\|f\|=1} |f(x)| \leq \|x\|$.

Let $f \in X^*$ with $\|f\| = 1$. Then by definition of operator norm,

$$|f(x)| \leq \|f\| \|x\| = \|x\|.$$

Taking supremum over all such f , we get

$$\sup_{\|f\|=1} |f(x)| \leq \|x\|.$$

Step 2: Show $\|x\| \leq \sup_{\|f\|=1} |f(x)|$.

If $x = 0$, the result is trivial. Assume $x \neq 0$.

By Corollary 1, there exists $f_0 \in X^*$ such that

$$\|f_0\| = 1 \quad \text{and} \quad f_0(x) = \|x\|.$$

Thus,

$$\sup_{\|f\|=1} |f(x)| \geq |f_0(x)| = \|x\|.$$

Step 3: Combine.

From Steps 1 and 2,

$$\sup_{\|f\|=1} |f(x)| = \|x\|.$$

Step 4: Equivalent formulation.

For any nonzero $f \in X^*$, define $g = \frac{f}{\|f\|}$, then $\|g\| = 1$, and

$$\frac{|f(x)|}{\|f\|} = |g(x)|.$$

Taking supremum over all $f \neq 0$ gives the equivalent form. □

3.3 Corollary 3A: Separation of Points

Corollary 3.3. *Let X be a normed linear space and let $x \in X$, $x \neq 0$. Then there exists $f \in X^*$ such that*

$$f(x) \neq 0.$$

Equivalently, if $x \neq y$, then there exists $f \in X^$ such that*

$$f(x) \neq f(y).$$

Proof. If $x \neq 0$, then by Corollary 1, there exists $f \in X^*$ such that

$$\|f\| = 1 \quad \text{and} \quad f(x) = \|x\| \neq 0.$$

For the second statement, if $x \neq y$, then $x - y \neq 0$, so there exists f such that

$$f(x - y) \neq 0 \implies f(x) \neq f(y).$$

□

3.4 Corollary 3B: Separation from a Closed Subspace

Corollary 3.4. *Let X be a normed linear space and $Y \subset X$ be a proper closed subspace. Let $x_0 \in X \setminus Y$ and define*

$$\delta := \text{dist}(x_0, Y) = \inf_{y \in Y} \|x_0 - y\| > 0.$$

Then there exists a continuous linear functional $f \in X^$ such that*

$$f|_Y = 0, \quad f(x_0) = \delta, \quad \|f\| = 1.$$

Proof. Define a subspace

$$M := Y + \text{span}\{x_0\}.$$

Every element of M can be uniquely written as $y + \alpha x_0$, where $y \in Y$, $\alpha \in \mathbb{R}$.

Define $f_0 : M \rightarrow \mathbb{R}$ by

$$f_0(y + \alpha x_0) = \alpha \delta.$$

Step 1: Well-definedness. If $y + \alpha x_0 = y' + \alpha' x_0$, then

$$(\alpha - \alpha')x_0 = y' - y \in Y.$$

If $\alpha \neq \alpha'$, this implies $x_0 \in Y$, a contradiction. Hence $\alpha = \alpha'$ and $y = y'$, so f_0 is well-defined.

Step 2: Linearity. Clearly f_0 is linear.

Step 3: Norm estimate. We claim that $|f_0(z)| \leq \|z\|$ for all $z \in M$.

Let $z = y + \alpha x_0$. Then

$$|f_0(z)| = |\alpha| \delta.$$

If $\alpha = 0$, trivial. If $\alpha \neq 0$, write

$$z = \alpha \left(x_0 + \frac{y}{\alpha} \right),$$

so

$$\|z\| = |\alpha| \left\| x_0 + \frac{y}{\alpha} \right\|.$$

Since $\frac{y}{\alpha} \in Y$,

$$\left\| x_0 + \frac{y}{\alpha} \right\| \geq \delta.$$

Thus,

$$\|z\| \geq |\alpha| \delta = |f_0(z)|.$$

Hence $\|f_0\| \leq 1$.

Step 4: Apply Hahn–Banach. Extend f_0 to $f : X \rightarrow \mathbb{R}$ such that

$$\|f\| = \|f_0\| \leq 1, \quad f|_M = f_0.$$

Step 5: Properties. For $y \in Y$,

$$f(y) = f_0(y) = 0.$$

Also,

$$f(x_0) = f_0(x_0) = \delta.$$

Finally,

$$\delta = f(x_0) \leq \|f\| \|x_0\| \implies \|f\| \geq \frac{\delta}{\|x_0\|}.$$

Combining with $\|f\| \leq 1$, we can normalize (if needed) to ensure $\|f\| = 1$.

Thus, the required functional exists. □

• Closed Graph Theo + Bounded inverse + Open mapping

X, Y NLS.

$A: X_0 \subseteq X \rightarrow Y$ is closed op
if for seq (x_n) s.t. $x_n \rightarrow x$ in X
 $x_n \in X_0$

& $Ax_n \rightarrow y$ then $Ax = y$ i.e. $x \in X_0$

• $G_A(A) = \{(x, Ax) \mid x \in X_0\}$ is closed
iff A is closed.

• If A is closed A may not be bounded.

• If A is bdd then it need not be closed.

Bdd + additional condition = closedness

Qn: closedness + additional condition $\Rightarrow$ bdd?

Closed Graph Theo

✓ If $\underline{X}$ and $\underline{Y}$ are Banach spaces
then every closed op $A: \underline{X} \rightarrow \underline{Y}$
is continuous.

Another version.

 If X & Y are B.S then
every closed op $A: X_0 \subseteq X \rightarrow Y$
is continuous if X_0 is closed.

Proof: Let X, Y be a B.S and
 $A: X \rightarrow Y$ is a closed op.

$$\text{Let } B_0 = \{x \in X \mid \|x\| < 1\}$$

We have to show

$$B_0 \subseteq \overline{\{x \in X \mid \|Ax\| \leq c\}} = V_c, \quad c > 0$$

$$\text{Let } V_\alpha = \{x \in X \mid \|Ax\| \leq \alpha\} \quad \alpha > 0.$$

$$\text{Claim } X = \bigcup_{j=1}^{\infty} V_j$$

$$V_j = \{x \in X \mid \|Ax\| \leq j\}.$$

Since X is a Banach space
 so by Baire-Catagory theorem
 interior of , some $k \in \mathbb{N}$, $\overline{V_k}$ is
 non-empty .

$\Rightarrow \exists x_0 \in X$ & $r > 0$ s.t

$$B(x_0, r) \subseteq \overline{V_k}$$

$\Rightarrow$
 [Let $\boxed{B_0 \subseteq \overline{V_{\frac{2k}{r}}}}$ $\xrightarrow{\neq \overline{W}}$ ① $\|x\| < 1$
 $x \in B_0$]

Then $u = \underline{x_0 + rx} \in \underline{B(x_0, r)} \subseteq \underline{\overline{V_k}}$

$\Rightarrow \exists u_n \in V_k$ s.t $u_n \rightarrow u$ as $n \rightarrow \infty$

$$\exists x_n \in V_k \text{ s.t. } x_n \rightarrow x_0 \text{ as } n \rightarrow \infty$$

$$[\because x_0 \in \overline{V_k}]$$

Then we have .

$$\underline{x} = \frac{1}{r} (u - x_0) = \lim_{n \rightarrow \infty} \frac{1}{r} (u_n - x_n)$$

Consider

$$\left\| A \left(\frac{u_n - x_n}{r} \right) \right\| \leq \frac{1}{r} (\|A u_n\| + \|A x_n\|)$$

$$\leq \frac{2k}{r}$$

$$\therefore \frac{u_n - x_n}{r} \in V_{\frac{2k}{r}} \Rightarrow \underline{x} \in \underline{\overline{V_{\frac{2k}{r}}}}$$

Denote $V_{\frac{2k}{r}}$ by W

Let $x \in B_0$, $0 < \epsilon < 1$.
 Since $\underline{B_0} \subseteq \underline{W}$ (by ①)

$$\exists x_1 \in W \quad \text{s.t.} \\ \|x - x_1\| < \epsilon$$

$$\Rightarrow \underbrace{\| \epsilon^{-1} (x - x_1) \|}_{\in B_0} < 1$$

$$\therefore \epsilon^{-1} (x - x_1) \in B_0$$

By the same argument $\exists x_2 \in W$
 s.t. $\| \epsilon^{-1} (x - x_1) - x_2 \| < \epsilon$

$$\| (x - x_1) - \epsilon x_2 \| < \epsilon^2$$

$$\Rightarrow \| x - (x_1 + \epsilon x_2) \| < \epsilon^2$$

Having obtained $x_1, x_2, \dots, x_n \in W$

$$\text{s.t. } \| x - (x_1 + \epsilon x_2 + \epsilon^2 x_3 + \dots + \epsilon^{n-1} x_n) \| < \epsilon^n$$

Since

$$\underline{B_0 \subset \overline{W}}$$

$$\| \epsilon^{-n} (x - (x_1 + \dots)) \| < 1.$$

So $\exists x_{n+1} \in W$ s.t

$$\| \epsilon^{-n} (x - (x_1 + \dots)) - x_{n+1} \| < \epsilon$$

$$\Rightarrow \| x - (\underbrace{x_1 + \epsilon x_2 + \dots + \epsilon^{n-1} x_n}_{\forall n}) \| < \epsilon^n \quad \text{--- (2)}$$

By induction $\exists (x_n)_{n \in \mathbb{N}}, x_n \in W$
s.t (2) holds $\forall n$.

$$\text{Let } s_n = \sum_{j=1}^n \epsilon^{j-1} x_j$$

$$\therefore \boxed{s_n \rightarrow x \text{ as } n \rightarrow \infty}.$$

Recall $x_j \in W \Rightarrow \|Ax_j\| \leq \frac{2k}{r}$

Then for $n > m$,

$$\begin{aligned}\|As_n - As_m\| &\leq \sum_{j=m+1}^n \|A(\epsilon^{j-1} x_j)\| \\ &\leq \frac{2k}{r} \sum_{j=m+1}^n \epsilon^{j-1}\end{aligned}$$

Then (As_n) is a Cauchy seq in Y .

Since Y is a Banach space

$\Rightarrow (As_n)$ is convergent and

converges to some $y \in Y$.

Given

A is a closed op.

Since $s_n \rightarrow x$ as $n \rightarrow \infty$

$$\& A s_n \rightarrow y, \quad y \in Y.$$

$$\therefore Ax = y \quad \text{--- (3)}$$

Consider $\|A s_n\| \leq \sum_{j=1}^n \|A(e^{j-1} x_j)\|$

$$\leq \frac{2K}{r} \sum_{j=1}^n e^{j-1} \leq \frac{2K}{r(1-\epsilon)}$$

$$\therefore \|Ax\| = \lim_{n \rightarrow \infty} \|A s_n\| \leq \frac{2K}{r(1-\epsilon)} = c \quad (\text{say})$$

$$\therefore x \in V_N$$

$$\Rightarrow B_0 \subseteq V_c, \quad c = \frac{2K}{r(1-\epsilon)}$$

~~$\Rightarrow$~~ ~~$\odot$~~ $\therefore A$ is continuous.

Theo. Let X, Y be Banach spaces.
Ex. $A \subseteq X_0 \subseteq X \rightarrow Y$ be a closed op.

Then A is continuous iff X_0 is closed in X .

[Recall.
Theo.

$A : X_0 \subseteq X \rightarrow Y$ be a bdd op.
• if X_0 is closed in X then A is closed.

• If Y is a B.S. & A is closed then X_0 is a closed subspace]

2 closed graph Theo.

Ex. ^{Assignment.}

Let $X = \ell^2$, (λ_n) be a seq
of positive scalars s.t $\lambda_n \rightarrow 0$
as $n \rightarrow \infty$ &

$$X_0 = \left\{ x \in \ell^2 : \sum_{j=1}^{\infty} \frac{|x(j)|^2}{\lambda_j^2} < \infty \right\}$$

Let $A : X_0 \subseteq X \rightarrow X$ defined
by ,

$$(Ax)(j) = \frac{x(j)}{\lambda_j} \quad j \in \mathbb{N} \text{ \& } x \in X.$$

Show that A is closed but
unbdd op. Give reasons.

The closed graph theo gives
a criteria for the
continuity of projection.

Recall

$P: X \rightarrow X$ is a

projection if $P^2 = P$.

If P is continuous Ex.

- $N(P)$ is closed.
 - $R(P)$ is closed.
- as well.
- $\left[\begin{array}{l} R(P) = N(I-P) \\ N(P) = R(I-P) \end{array} \right]$

Cor.

2

Let X be Banach space
 $P: X \rightarrow X$ be a projection.

If $N(P)$ & $R(P)$ are closed
subspaces of X then P is
continuous.

Proof. To use closed graph
- theo it is enough to
show that P is closed.

Let $(x_n)_{n \in \mathbb{N}}$ be a seq in X
s.t. $x_n \rightarrow x$ and $Px_n \rightarrow y$
in X
then we have to show

$Px = y$.
Now since $R(P)$ is closed.

$$\Rightarrow y \in R(P) \Rightarrow Py = y.$$

Consider

$$x_n - Px_n \in R(I-P) = N(P)$$

$$x_n - Px_n \rightarrow x - y \quad (\text{given})$$

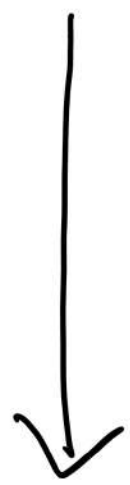
$$\text{as } n \rightarrow \infty$$

$\therefore x - y \in N(P)$ ~~2~~ as $N(P)$ is closed.

$$Px = y.$$

A consequence of closed graph
Theo.

Functions. : $A, B (\neq \emptyset)$ &
 $f: A \rightarrow B$ be a fn.



Then $f^{-1}: B \rightarrow A$ is well defined

$\Leftrightarrow f: A \rightarrow B$ is a bijection.

Vector space: X, Y linear space.

$T: X \rightarrow Y$ be a linear transformation.

Suppose $T: X \rightarrow Y$ is bijective.

then $T^{-1}: Y \rightarrow X$ is well defined.
it is linear (automatically)



N.L.S. : X, Y N.L.S.

$T: X \rightarrow Y$ bdd linear op.

Inverse : $T^{-1}: Y \rightarrow X$ we need.

$T: X \rightarrow Y$ bijective

Qn! Is $T^{-1}: Y \rightarrow X$ is a Bdd
op whenever $T: X \rightarrow Y$
is bdd linear
bijective map?

Ex. $X = (c_{00}, \|\cdot\|_1)$
 $Y = (c_{00}, \|\cdot\|_\infty)$

$T: X \rightarrow Y$ given by

$$Tx = x$$

• T is linear.

$$\begin{aligned} \bullet \quad |Tx(i)| &= |x(i)| \\ &\leq \sum_{i=1}^{\infty} |x(i)| = \|x\|_1, \end{aligned}$$

$$\therefore \|Tx\|_\infty \leq \|x\|_1$$

$\therefore T$ is bdd.

Now $T^{-1}: (C_{\infty}, \|x\|_{\infty}) \rightarrow (C_{\infty}, \|\cdot\|_1)$
is well defined. (check *)

$$T^{-1}(x) = x \quad \forall x \in C_{\infty}.$$

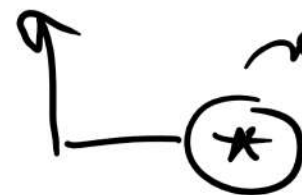
Claim: T^{-1} is not bdd ?

Suppose T^{-1} is bdd.

Then $\exists M > 0$ s.t

$$\|T^{-1}(x)\|_1 \leq M \|x\|_{\infty} \quad \forall x \in C_{\infty}.$$

Take a seq



$$x_n = (1, \dots, 1, 0, \dots, 0)$$

Then $x_n \in C_{00} \quad \forall n \in \mathbb{N}$.

$$\|x_n\|_{\infty} = 1$$

$$\|x_n\|_1 = n.$$

By $\textcircled{*}$

$$\|x_n\|_1 \leq M \|x_n\|_{\infty}.$$

$$\Rightarrow n \leq M \quad \forall n \quad \Downarrow$$

Note. X & Y are not Banach.

Theo. (Bounded Inverse Theo).

Let X, Y be B. Spaces &

$A: X_0 \subseteq X \rightarrow Y$ be an ~~op~~
injective closed op. Then

$A^{-1}: R(A) \rightarrow X$ is continuous
(bdd)

if $R(A)$ is a closed subspace.
in Y .

($\Leftarrow$ closed graph.
theo.)

Cor. Let X & Y be B.S.

$A \in B(X, Y)$ be bijective

Then $A^{-1} \in B(Y, X)$. $X_0 = X$.

- Ω_1 & Ω_2 be two metric spaces.

$$\underline{f}: \Omega_1 \rightarrow \Omega_2,$$

- Any bijjective bdd lin op between two Banach spaces is linear homeomorphism.

Open Mapping Theo.

$$\Omega_1, \Omega_2$$

$$f: \Omega_1 \rightarrow \Omega_2 \text{ open map}$$

if for every $E \subseteq_{\text{open}} \Omega_1 \Rightarrow f(E)$ open in Ω_2

- $f: \Omega_1 \xrightarrow{\text{open}} \Omega_2 \quad g: \Omega_2 \xrightarrow{\text{open}} \Omega_3$

Then $f \circ g$ is open.

- $f: \Omega_1 \rightarrow \Omega_2$ is bijective + open
 $\Rightarrow f^{-1}$ is continuous.

Prop. Let X & Y be two normed
linear spaces & $A: X \rightarrow Y$ linear
operator. If A is an open map
then A is surjective.

Proof: $A: X \rightarrow Y$ open.

Then $A(X)$ is open since X is
open.

- If $A(X) \subsetneq Y$, since Y is a nls
 $\overline{A(X)}$ will have empty
interior.

$\downarrow$

$\therefore A(X) = Y$.

Open map $\Rightarrow$ surjectivity.

Converse?

Open mapping theo.

Lemma.

Let Z be a closed subspace of X (nls). Then $h: X \rightarrow X/Z$ is linear, surjective & continuous.
+ open map.

Proof.

To show h is continuous.

$$\bullet \quad \|h(x)\| = \|x + Z\| \leq \|x\|$$

$\therefore h$ is continuous.

$$\begin{array}{l} Z \subseteq X \\ X/Z \\ h: X \rightarrow X/Z \\ h(x) = x + Z \end{array}$$

Let $G \subseteq X$. To show $\eta(G)$ is open,
 $\downarrow$ open

for $\underline{x_0} \in G$ $\exists x \in X$ s.t. for
some $r > 0$

$$\|x - x_0\| < r \Rightarrow \underbrace{B(x_0, r) \subseteq G}_{(\Rightarrow x \in G)}$$

To show,

$$\|(x+z) - (\underline{x_0} + z)\| < r.$$

$$\left[B(x_0 + z, r) \subseteq \eta(G) \right]$$

$$\Rightarrow x + z \in \eta(G)$$

Now for $x \in X$ with

$$\|x + z - x_0 + z\| < r$$

$$\Rightarrow \exists z \in Z \text{ s.t. } \|x - x_0 + z\| < r \\ \Rightarrow \|(x+z) - x_0\| < r$$

$$\Rightarrow x+z \in G.$$

$$\begin{aligned} \Rightarrow h(x+z) &= (x+z) + Z \\ &= x + Z \in \eta(G) \end{aligned}$$

- Quotient map is linear, surjective, continuous & open.

$$\begin{aligned} A: X &\rightarrow Y \quad (\text{linear operation}) \\ Z &\subseteq X \\ &\text{closed.} \end{aligned} \quad X/Z \rightarrow \text{nl's.}$$

$$\tilde{A}: X/Z \rightarrow Y,$$

$$\tilde{A}(x+z) = Ax \quad \text{if } Z \subseteq N(A).$$

- $\tilde{A}$ is linear.

• $\tilde{A}$ is well defined.

Lemma. X, Y n.l.s.

$\tilde{A} \in B(X, Y)$. $\& \quad Z \subseteq X \quad \& \quad Z \subseteq N(A)$
closed

Then $\tilde{A}$ is a bdd linear
op $\& \quad \|\tilde{A}\| = \|A\|$.

Proof. To show, $\|\tilde{A}(x+z)\| \leq M\|x+z\|$

$$\begin{aligned} \|\tilde{A}(x+z)\| &= \|Ax\| = \|A(x+u)\| \\ &\leq \|A\| \|x+u\| \quad \forall u \in Z. \\ &\leq \|A\| \inf_{u \in Z} \|x+u\| \quad \forall u \in Z \quad \text{ } \overset{\subseteq N(A)}{\end{aligned}$$

$$= \|A\| \|x+z\|$$

$\therefore \hat{A}$ is bdd.

$$\|\hat{A}\| \leq \|A\| \text{ --- (1)}$$

$$\|A(x)\| = \|\hat{A}(x+z)\|$$

$$\leq \|\hat{A}\| \|x+z\| \leq \|\hat{A}\| \|x\|$$

$$\therefore \|A\| \leq \|\hat{A}\| \text{ --- (2)}$$

$$\therefore \|A\| = \|\hat{A}\|.$$

Open mapping.

Let X, Y be Banach spaces & $A \in B(X, Y)$ then
if A is surjective $\Rightarrow A$ is open.

Proof! Since A is bdd
 $\Rightarrow N(A)$ is closed.

Take $Z = N(A)$.

Then $\tilde{A} : X/Z \rightarrow Y$ is bdd.

Claim. $\tilde{A}$ is injective.

$$\tilde{A}(x+z) = \tilde{A}(y+z)$$

$$\Rightarrow Ax = Ay$$

$$\Rightarrow x - y \in N(A) = Z$$

$$\Rightarrow x - y + z = z$$

$$\Rightarrow x + z = y + z$$

$\therefore \tilde{A}$ is injective.

Claim: $\tilde{A}$ is surjective.

Since A is surjective so for
any $y \in Y \exists x$ s.t

$$Ax = y.$$

$$\therefore y = \tilde{A}(x+z)$$

Thus $\tilde{A}$ is a bijective map
+ bdd.

$X/Z \subseteq Y$ Banach space.

Then by B.I.T we have,
 $\tilde{A}^{-1}$ is bdd (continuous)

$\Rightarrow \tilde{A}$ is open. $\left[\begin{array}{l} U \subseteq X/Z \text{ open.} \\ \Rightarrow \tilde{A}(U) \underset{\text{open}}{\subseteq} Y \\ \tilde{A}(U) = (\tilde{A}^{-1})^{-1}(U) \end{array} \right]$

Observe —

$$\eta: X \rightarrow X/Z$$

$$\tilde{A}: X/Z \rightarrow Y$$

$$\tilde{A}(x+z) = Ax$$

$$\therefore \tilde{A} \circ \eta = A$$

$\therefore A$ is an open map.

Ex. Every non-zero continuous linear fnal on a B.S is an open map.

[Hint show $\tilde{A}$ fnal is surjective].

Uniform & Bdd Principle.

Qn. X, Y nls.
- (A_n) seq of linear operators.
s.t $(A_n x)_{n \in \mathbb{N}}$ converges for each
 $x \in X$.

Define $A: X \rightarrow Y$

$$A(x) = \lim_{n \rightarrow \infty} A_n x.$$

• A linear.

Qn: Is A bdd?

Suppose. $A_n \in B(X, Y)$ s.t $\{\|A_n\|\}$
is bdd. Then $A: X \rightarrow Y$
is bdd, $\|A\| \leq \liminf \|A_n\|$

Given $Ax = \lim_{n \rightarrow \infty} A_n x.$

$$\begin{aligned} \therefore \|Ax\| &= \lim_{n \rightarrow \infty} \|A_n x\|. \\ &= \liminf_{n \rightarrow \infty} \|A_n x\| \\ &\leq \liminf \|A_n\| \|x\|. \end{aligned}$$

$$\therefore \|A\| \leq \liminf \|A_n\|.$$

$$A_n \in \overline{B(x, y)}, \quad \{\|A_n\|\} \text{ bdd.}$$

$$\Rightarrow A \text{ is bdd.}$$

If. we reduce $A_n \in B(x, y).$

$$\Rightarrow A \text{ is bdd.}$$

Ex. $X = c_0$, $\|\cdot\|_\infty$, $Y = \mathbb{K}$.

$$\underline{f_n(x)} = \sum_{j=1}^n x(j), \quad x \in X.$$

for each n
 $\|f_n\| = n$.

$$\boxed{\begin{aligned} |f_n(x)| &\leq \|x\|_\infty n \\ \therefore \|f_n\| &\leq n. \end{aligned}}$$

$$\underline{\underline{f_n}} \in B(X, \mathbb{K})$$

$$f_n(x) \rightarrow \sum_{j=1}^{\infty} x(j) = \underline{f}(x)$$

But f is not continuous.

$$\begin{aligned} x_n(j) &= 1, & j \leq n \\ &= 0, & j > n. \end{aligned}$$

$$\|x_n\|_\infty = 1 \quad |f(x_n)| = n.$$

$\therefore f$ is not continuous.

Defn. Let X, Y be NLS
 $\mathcal{E} \subseteq X$.

A family $\mathcal{A}$ of linear
operators from $X \rightarrow Y$ is said
to be pointwise bounded on $\mathcal{E}$
if for each $x \in \mathcal{E} \exists M_x > 0$
s.t. $\|Ax\| \leq M_x, \quad \underline{\forall A \in \mathcal{A}}.$

Further $\mathcal{A}$ is said to be
uniformly bdd. if $\exists M > 0$
s.t. $\|Ax\| \leq M_x \quad \forall x, \forall A.$

Q: Let $X = C[a, b]$ with sup norm.
and $Y \subseteq X$, Y consisting of all
constant fns.

Let $g \in Y'$ defined by

$$g(y) = y(a) \quad \forall y \in Y.$$

Define $f(x) = \frac{1}{b-a} \int_a^b x(t) dt$,
 $x \in X$.

Then show that f is a Hahn-Banach extension of g .